SOFTWARE PROJECT LAB 1

ARGUS

EXAM SUPERVISION SYSTEM

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PROJECT DESCRIPTION

Argus is a real-time exam supervision system designed to ensure academic integrity by monitoring student network activity.

It captures DNS packets using **Raw Sockets**, filters them against a strict whitelist, and securely reports violations to a centralized supervisor dashboard, automating the proctoring process for large-scale examinations.

MOTIVATION

ACADEMIC INTEGRITY

Prevents digital cheating by blocking unauthorized resources.

AUTOMATION

Reduces the manual burden on proctors through automated alerts.

SCALABILITY

Capable of monitoring hundreds of students simultaneously.

SYSTEM WORKFLOW

PACKET CAPTURE

Intercepts DNS traffic on using Raw Sockets to extract requested domain names.

LOCAL FILTER

Verifies domains against a local Hash Table (Whitelist) and Trie (Wildcards) to detect violations.

ENCRYPT PROTOCOL

Serializes violation data, encrypts the payload via XOR, and appends a checksum for integrity.

SERVER LOG

Multiplexes multiple clients via select(), decrypts alerts, and updates the real-time Dashboard.

CORE TECHNOLOGY

C++

High-performance backend logic and memory management.

RAW SOCKETS

Low-level packet sniffing for traffic analysis.

GIT

Distributed version control for source code management.

GITHUB

Collaboration platform for code review and hosting.

KEY DATA STRUCTURES

HASH TABLE

O(1) lookup speed for exact domain whitelist matching.

TRIE

Efficient prefix matching for wildcard domains (e.g., *.edu).

PRIORITY QUEUE

Min-heap to rank top student violators in real-time.

CURRENT PROGRESS

PROJECT CONFIGURATION

Modular system architecture set up with full configuration integration.

✓COMPLETE

NETWORKING CORE

Robust Multi-client TCP Server and Echo Client communication established.

✓COMPLETE

DATA STRUCTURES

Implemented Hash Tables for whitelisting and Tries for wildcard matching.

✓COMPLETE

SECURE PROTOCOL

Custom communication protocol with XOR encryption and checksums active.

✓COMPLETE

FUTURE PLAN

DNS CAPTURE

Implement DNS Packet Capture using raw sockets for real-time traffic analysis.

NEXT STEP

SERVER DASHBOARD

Develop a real-time terminal UI to visualize violations and logs instantly.

PLANNED

SYSTEM RESILIENCE

Add offline message queuing to handle network drops without data loss.

PLANNED

ADVANCED SECURITY

Integrate anti-tampering mechanisms and time verification checks.

PLANNED

Thank You

Github Repository : https://github.com/DevNexus09/Argus-ExamSupervisonSystem